BRINGFY DELIVERY INC.

Technical Whitepaper

Proprietary Software & Algorithmic Innovation

Last-mile Delivery Logistics Technology Platform
Business Technical Documentation

Company	Bringfy Delivery Incorporated.
Website	bringfy.ca
Incubator	Brilliant Catalyst — Ontario Tech University
Document Type	Technical Whitepaper — Proprietary Software & IP
Prepared By	Olavo Mello - CTO | Priscila Calil - COO | Vinicius Medeiros - CEO
Date	April 2026
Classification	Confidential

1. EXECUTIVE SUMMARY

Bringfy Inc. is a Canadian-incorporated logistics technology company (LogTech) that has developed a proprietary, cloud-based last-mile delivery management platform. The platform connects small and medium-sized businesses (SMBs) with professional couriers and drivers through an intelligent, algorithm-driven dispatch system.

This whitepaper describes the technical architecture, proprietary algorithms, and unique innovations embedded in the Bringfy platform. It demonstrates that the software constitutes original, novel intellectual property developed by the Bringfy founding team — specifically by CTO Olavo Mello, with over 26 years of software development experience — and that this technology represents a meaningful and differentiated contribution to the Canadian logistics technology market.

The Bringfy platform has been validated through live product deployment at panel.bringfy.ca, beta testing with Canadian businesses, and recognition from major industry bodies. The company operates from Canada and is supported by Brilliant Catalyst at Ontario Tech University under the Startup Visa Program.

2. PROBLEM STATEMENT

Last-mile delivery — the final leg of a supply chain, from a distribution point to the end customer — represents the most expensive and operationally complex segment of logistics. For SMBs in Canadian urban and suburban markets, this challenge is particularly acute:

- Existing enterprise logistics platforms (e.g., Onfleet) require companies to maintain their own private fleet, which is cost-prohibitive for SMBs.
- Marketplace-only platforms (e.g., Shippie, GoBolt) assign deliveries to their own fleets, denying businesses control and flexibility over their logistics operations.
- No platform on the Canadian market allows a business to simultaneously manage its own existing fleet AND access an on-demand network of partner drivers, with automated dispatch and real-time load optimization.
- Pricing in existing solutions is opaque and static, disadvantaging smaller operators who cannot negotiate volume rates.

Bringfy was designed specifically to solve these gaps, enabling any business — regardless of size — to access a professional, automated, and cost-competitive delivery infrastructure.

3. PLATFORM ARCHITECTURE

The Bringfy system is composed of four tightly integrated proprietary modules, all developed in-house by the founding technical team:

3.1 B2B Management Panel (panel.bringfy.ca)

A cloud-hosted, web-based back-office platform that gives business clients full control over their delivery operations. Key features include:

- Real-time delivery request creation with cargo type classification (item, furniture, oversized)
- Automated carrier matching based on geolocation, vehicle load capacity, and availability
- Live order tracking dashboard with map visualization and status timeline
- Delivery history, invoice management, and payment processing via Stripe integration
- QR-code generation per order for scan-based delivery confirmation

3.2 Client Mobile Application

A cross-platform mobile application enabling end-customers to request deliveries directly. Features include streamlined order placement, real-time driver tracking, push notifications, and delivery confirmation with photographic evidence capture.

3.3 Driver Application

A purpose-built application for carrier partners that manages the driver-side of the delivery workflow. The app handles:

- Authentication and onboarding (Google, Facebook, Apple, email)

- Real-time delivery opportunity push notifications based on geolocation proximity
- Route optimization with integrated map navigation
- Load specification display (dimensions, weight, cubic weight)
- QR code scanner for pickup and delivery confirmation
- Transport report with photographic documentation
- Commission and earnings tracking

3.4 Cloud Infrastructure

The platform is hosted on Amazon Web Services (AWS), migrated from initial infrastructure in September 2023 following a USD $5,000 AWS for Startups grant approval. The system uses a scalable microservices architecture designed to support multi-region expansion. Database management, API routing, and load balancing are handled via AWS services.

4. PROPRIETARY ALGORITHMS & UNIQUE INNOVATIONS

The core of Bringfy's technological differentiation lies in four proprietary algorithmic systems developed by the founding team. These algorithms form the intellectual core of the platform and are not replicated by any competitor identified in the Canadian last-mile market.

4.1 Dynamic Geospatial Dispatch Algorithm

Bringfy's central algorithm continuously processes real-time data inputs to automatically match delivery requests with the most suitable available driver. The matching logic evaluates:

Input Variable	Description
Driver Geolocation	Real-time GPS coordinates of all active partner drivers
Pickup Proximity	Haversine distance calculation between driver and pickup point
Vehicle Load Capacity	Carrier vehicle class matched against cargo dimensions and cubic weight
Driver Availability Status	Live availability flag set by driver in the partner app
FIFO Queue Priority	First-In-First-Out ordering within matching-tier to ensure fairness
Time Window Compatibility	Scheduled pickup time vs. driver current route completion estimate

The algorithm applies a multi-tier scoring function. Candidates are ranked by a weighted composite of proximity, capacity match, and availability. The top-ranked driver receives a push notification; if they do not accept within a configurable timeout window, the next-ranked candidate is notified. This avoids the idle-time losses common in manual dispatch systems.

4.2 Automated Cubic Weight & Pricing Engine

Bringfy is uniquely the only Canadian last-mile logistics platform that generates delivery pricing automatically and standardized at the point of order creation, without requiring manual quotation or negotiation. The pricing engine works as follows:

- The client inputs cargo dimensions (height, width, length in cm) and weight (grams).
- The system computes cubic weight using the standard formula: (H × W × L) / 5,000.
- The final chargeable weight is the greater of actual weight and cubic weight (industry-standard dimensional weight pricing).
- A base fare is applied from a configurable rate matrix, adjusted by distance zone, vehicle class required, and time-of-day factors.
- All pricing is generated in real time, displayed to the client before order placement, and locked upon confirmation.

This automated pricing transparency is a significant market differentiator. Competing platforms require manual quotes or post-delivery billing, creating friction and unpredictability for SMB clients.

4.3 Idle Fleet Utilization & Capacity Optimization

A critical innovation in the Bringfy platform is its ability to activate idle, under-utilized fleet capacity — a problem that persists across the logistics industry globally. The system:

- Maintains a registered pool of partner drivers (independent couriers, part-time workers, private fleet owners) who opt in to receive delivery requests when available.
- Monitors declared availability windows and cross-references them with incoming order demand in real time.
- Dynamically adjusts driver pool prioritization based on historical acceptance rates, delivery performance scores, and geographic distribution patterns.
- Enables businesses with their own fleet to integrate their drivers directly into the platform — a hybrid model in Canadian competitor currently offers.

This approach directly addresses the economic and environmental costs of empty-leg driving and underutilized fleet assets, contributing to both cost reduction for clients and lower carbon footprint per delivery — a key sustainability feature of the platform.

4.4 Real-Time Status & Delivery Intelligence System

The platform maintains a live operational state machine for every active delivery order, tracking transitions across defined states:

Order Created → Payment Pending → Confirmed → Driver Assigned → Collecting → In Transit → Delivered → Completed

Each state transition is timestamped and logged, creating a complete audit trail. The system generates automatic notifications to both the business client and end-recipient at key transition points. The photographic confirmation at delivery (captured via the Driver App) creates verifiable proof of delivery with geolocation tagging.

5. COMPETITIVE DIFFERENTIATION

Feature	Bringfy	GoBolt	Onfleet	Shippie / Tyltgo

Hybrid own fleet + partner network	✓	✗	✗	✗
Automated real-time pricing	✓	✗	✗	✗
B2B management panel	✓	✗	✓	✗
Driver mobile app	✓	✓	✓	✓
Client mobile app	✓	✗	✗	✓
Cubic weight pricing engine	✓	✗	✗	✗
FIFO dispatch algorithm	✓	✗	✗	✗
SMB-focused pricing	✓	✗	✗	Partial
Donation delivery module	✓	✗	✗	✗

6. INTELLECTUAL PROPERTY OWNERSHIP

All software, algorithms, and systems described in this whitepaper were designed and developed exclusively by the Bringfy Inc. founding team:

- Olavo Mello (CTO) — Architecture, full-stack development, platform engineering. 26+ years of professional software development, project management, and digital marketing experience.
- Vinicius Medeiros (CEO/CFO) — System concept origination, georeferencing methodology, logistics domain logic. 12+ years in satellite tracking and cargo management.
- Priscila Calil (COO) — Transport operations requirements, regulatory compliance specifications. 12+ years in transport infrastructure.

All intellectual property, source code, algorithms, database schemas, user interface designs, and product documentation are owned exclusively by Bringfy Inc., a corporation registered in Canada. No third-party proprietary algorithms or licensed platform components form the core dispatch or pricing logic of the platform.

IP Component	Ownership & Status
Platform Source Code	Proprietary — Bringfy Inc. (copyright automatic under Canadian law)
Dispatch Algorithm	Proprietary — Trade secret, internally documented
Pricing Engine Logic	Proprietary — Trade secret, internally documented
B2B Management Panel	Proprietary — bringfy.ca / panel.bringfy.ca
Database Architecture	Proprietary — Self-hosted, Bringfy Inc.
UX/UI Design System	Proprietary — Developed by Bringfy design team

7. PLATFORM VALIDATION & MARKET TRACTION

The Bringfy platform is not a concept — it is a functioning, deployed, and validated product. The following milestones demonstrate real-world operation and market proof:

7.1 Brazil Operations (e-Carreto)

The predecessor platform, e-Carreto (ecarreto.com.br), launched in Brazil in January 2022 and within 5 months achieved over 10,000 orders, 4,000 registered users, and 1,000+ active drivers. This real-world deployment validated the core dispatch and matching algorithms at scale under live market conditions.

7.2 Canadian Platform Development & Deployment

Date	Milestone
May 2023	B2B Panel prototype launched at panel.bringfy.ca
June 2023	Client App MVP and Driver App MVP completed
July 2023	B2B / B2C integration completed; partnership with SELC College
August 2023	B2B Panel officially launched; Communitech membership
September 2023	AWS for Startups grant (USD $5,000) approved; AWS migration
November 2023	Microsoft Sponsorship (USD $150,000) received
November 2023	Riipen ICT Ignite+ program — cybersecurity integration
January 2024	First Canadian B2B client meeting (early adopter)
February 2024	Beta test: moving service (driver: Rafael)
April 2024	Beta test: moving service (Marina — YMCA)
May 2024	Recognized: Top 10 Logistics Providers in Latin America — Logistics & Transportation Review LATAM
December 2024	Joint venture signed: DeliveryIt + Bringfy + e-Carreto
April 2025	BlackBerry Development Partner — enterprise security integration
October 2025	Beta test: product delivery with JJ's Convenience & Café
January 2026	Recognized: Top Logistics Network Platform in Latin America 2026 — Logistics Transportation Review

7.3 Third-Party Recognition & External Validation

- Incubated by Brilliant Catalyst at Ontario Tech University — confirmed in writing by Executive Director Osman Hamid (October 2025)
- Attended Collision Conference 2023 and 2024 as part of Brilliant Catalyst delegation — resulting in international market development inquiries from Business France (French Embassy in Canada)
- Invitation from Home Delivery World 2026 (Nashville) to participate in the Start-Up Village — recognition by one of North America's premier logistics and supply chain conferences
- Business development contact from Mellohawk Logistics (Mississauga, ON) — a Transport Canada Regulated Agent — for B2B trade calculation collaboration

- Mentorship and commercial guidance from Bluekey Consulting; legal IP advisory from Torys LLP; financial planning with BDO

8. ENVIRONMENTAL & SOCIAL INNOVATION

The Bringfy platform incorporates sustainability as a design principle, not an afterthought. Two specific innovations address environmental impact:

8.1 Idle Fleet Activation & Carbon Footprint Reduction

By routing deliveries through drivers who are already in proximity to a pickup location — rather than dispatching dedicated vehicles from a central depot — the platform significantly reduces the number of vehicle kilometers traveled per delivery. The FIFO-based dispatch algorithm inherently optimizes for proximity, reducing fuel consumption and emissions per order.

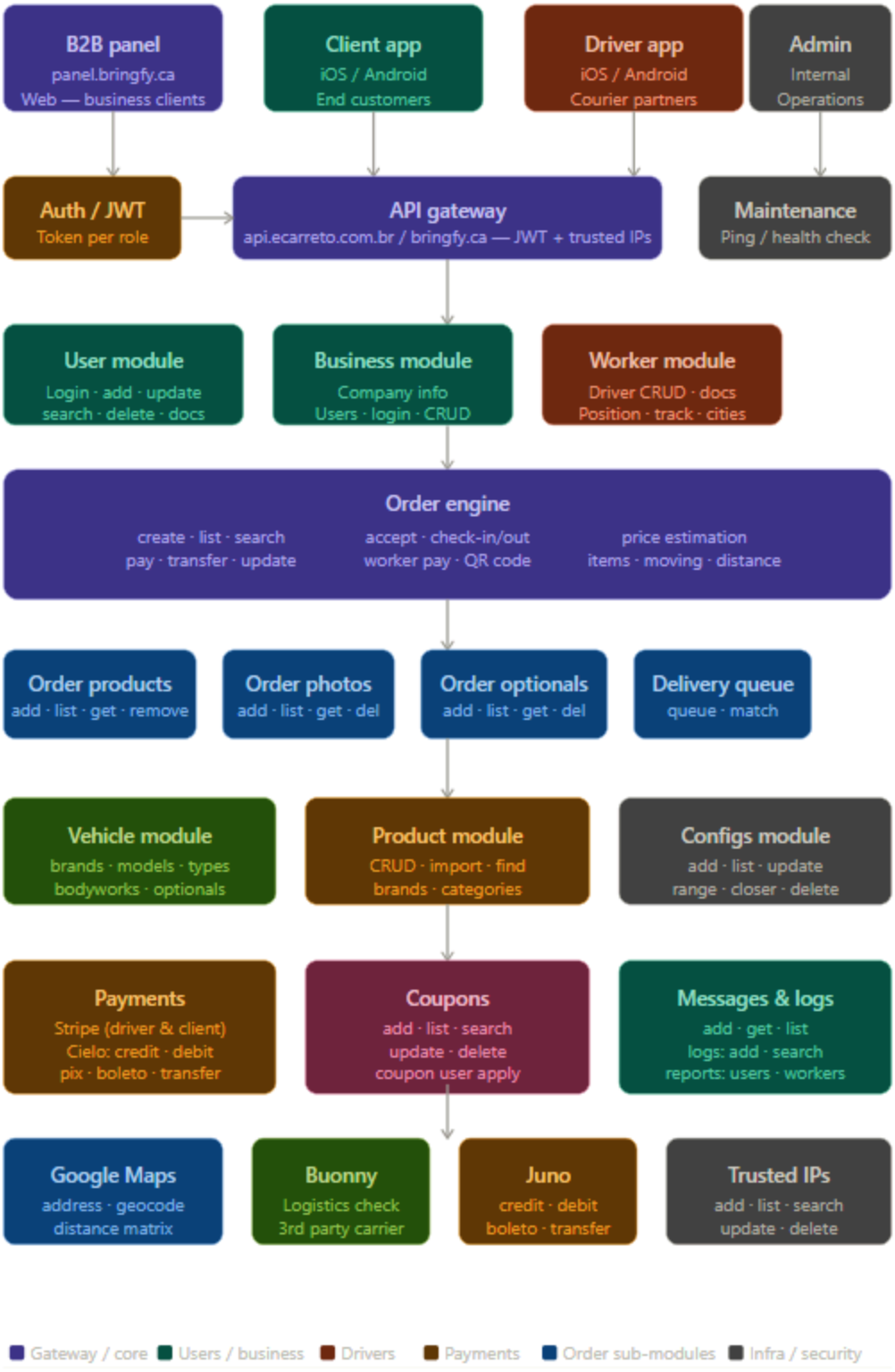
8.2 Donations Delivery Module

Bringfy has built a dedicated module within its platform for facilitating community donations. The system connects donors wishing to contribute goods with recipients and charitable organizations, using the same logistics infrastructure to coordinate pickup and delivery. This initiative has been cited by Brilliant Catalyst as evidence of Bringfy's social impact leadership in Canada.

9. FOUNDING TEAM & TECHNICAL CREDENTIALS

Name / Role	Technical & Professional Background
Olavo Mello — CTO	26+ years in software development, project management, startups, and digital marketing. Senior full-stack developer. Architect of the Bringfy platform and all proprietary algorithms.
Vinicius Medeiros — CEO/CFO	12+ years in satellite tracking, vehicle management, and cargo logistics. Originator of the georeferencing-based dispatch concept. Entrepreneur.
Priscila Calil — COO	12+ years in transport infrastructure. Heavy construction estimator and road infrastructure project manager. Defines operational and regulatory requirements for the platform.

10. SYSTEM API DIAGRAM



11. CONCLUSION

Bringfy Inc. has developed a proprietary, multi-component logistics technology platform that represents a genuinely novel contribution to the Canadian last-mile delivery market. The platform's

core innovations — its geospatial dispatch algorithm, automated cubic weight pricing engine, hybrid fleet management system, and real-time state machine — are not replicated by any competitor in the Canadian market.

The software has been validated through real-world deployment, beta testing with Canadian businesses, and external recognition from industry organizations in Canada, the United States, and internationally. All intellectual property is owned by Bringfy Inc. and developed by its founding team based in Canada.

This whitepaper, in conjunction with the supporting business evidence submitted as part of the Canada Startup Visa application, demonstrates that Bringfy Inc. meets and exceeds the criteria for innovative, scalable, and economically beneficial startup activity in Canada.